The Tom Klootwijk Manifold - Compact Literal Definitions v1.1

Public-sharing card - direct identifiers masked - TKM-STSDF-U63-A36-20260822-072511

A. Original 7-dimensional core

$$\boxed{\mathcal{K}_{\mathrm{TK}}(\mathbf{r}) = \prod_{i=0}^{6} S^1(r_i)}$$

$$\boxed{\mathcal{K}_{\mathrm{TK}}(\mathbf{r}) = \{x \in \mathbb{R}^{14} : x_{2i}^2 + x_{2i+1}^2 = r_i^2,\ i = 0, \dots, 6\}}$$

$$\Phi(\theta)_{2i:2i+1} = r_i(\cos\theta_i, \sin\theta_i), \quad g = \sum_{i=0}^{6} r_i^2 d\theta_i^2, \quad g(E_i, E_j) = \delta_{ij}.$$

Result: a smooth compact connected 7-torus with seven pairwise orthonormal tangent directions.

B. Spatiotemporal tubular SDF addendum

Let $t \in I$, $Q(t) \in SO(14)$, $c(t) \in \mathbb{R}^{14}$, $r_i(t) > 0$, and

$$0 < \tau(t) < r_* := \inf_{i,t} r_i(t).$$

Set $y = Q(t)^{\mathsf{T}}(x - c(t))$ and

$$\boxed{\Delta_i(x,t) = \|y_{2i:2i+1}\|_2 - r_i(t)}, \qquad \boxed{\mathcal{D}_\tau(x,t) = \|\Delta(x,t)\|_2 - \tau(t)}.$$

$$\boxed{\Delta^{-1}(0) \cong \mathbb{T}^7 \times I}, \qquad \boxed{\mathcal{D}_\tau^{-1}(0) \cong \mathbb{T}^7 \times S^6 \times I}.$$

$$\boxed{\|\nabla_x \mathcal{D}_\tau\|_2 = 1}, \qquad \boxed{V_n = -\partial_t \mathcal{D}_\tau}.$$

C. The compact substrate triple

$$\boxed{\mathfrak{S}_{\mathrm{TK}}^{\mathrm{ST}} = \left(\Delta^{-1}(0),\ \mathcal{D}_\tau^{-1}(0),\ \{\widehat{\mathcal{D}}_{\tau h}\}_{h>0}\right)}$$

$$\boxed{\|\widehat{\mathcal{D}}_{\tau h} - \mathcal{D}_\tau\|_{L^\infty(B)} = o(1) \quad (h \downarrow 0)}.$$

Here lowercase $o(1)$ means **vanishing numerical error**. It does not mean constant-time execution.

D. Complexity and rendering truth

One exact field query	Seven pair lengths plus one 7-vector length: fixed $O(7) = O(1)$.
Raymarched image	$O(PS)$ field queries for P pixels and at most S steps.
Memory	Nonzero code, render targets, uniforms, and engine resources.
Unity visual	Exact $\mathbb{R}^{14}$ API plus an exact 3D ring-torus SDF witness; the witness is not the entire high-dimensional shell.

E. Formal guard and failure boundary

Topology remains $\mathbb{T}^7 \times S^6$ while the data are smooth and $0 < \tau < \min_i r_i$. A topology/smoothness event requires a guard failure such as $r_i = 0$, $\tau = 0$, $\tau \geq \min_i r_i$, discontinuity, or a singular zero set.

Public occasion record

Tom Klootwijk; age 36; BSN NL*********; birth date **-**-1990; 2026-08-22T07:25:11+02:00; Unity 6000.3 LTS / URP 17.3.0; package 1.1.0.

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